

Heart Beat Perception and Emotional Experience

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ABSTRACT

The main assumption of the present study is that emotional experience is coupled to perception of bodily processes. From this it is deduced that individuals who show good perception of heart activity tend to exhibit higher levels of a momentarily experienced emotion (in this case anxiety) and to score higher on the personality trait "Emotional Lability." The aspect of cardiac awareness considered here is perception of heart beats. Subjects were instructed to count heart beats (only by concentrating on their body but not by taking their pulse) during a signalled time interval and the reported number of beats was then compared to the actual number of beats as extracted from the EKG. Additionally, skin conductance and respiration were recorded. During the experimental task (i.e., counting heart beats) increases in heart rate, respiration rate and number of spontaneous fluctuations in skin conductance were recorded. Groups of good and poor perceivers were formed on the basis of accuracy on the perception task. The two groups did not differ in heart rate; however, in State Anxiety and Emotional Lability, the group of good perceivers had significantly higher scores. The importance of visceroperception for emotional experience is pointed out and the relevance for clinical psychology is discussed.

DESCRIPTORS: Cardiac perception, Emotional experience, State Anxiety, Emotional lability.

The assumption that the perception of "bodily" functioning or "bodily" changes (visceral perception) is an accompanying element of emotional experience has a long history and seems to be widely accepted. Since James (1884, p. 189) stated that "... our feelings of the (same) changes [in body functions] as they occur, is the emotion," several studies have been undertaken in order to investigate the relationship of visceral perception to emotional experience. Mandler, Mandler, and Uviller (1958) approached this issue by studying the relationship between anxiety (as measured by the Manifest Anxiety Scale, MAS) and *reported* autonomic awareness, which was quantified by means of the Autonomic Perception Questionnaire (APQ). They obtained significant positive correlations between these two measures and also found higher autonomic reactivity for subjects scoring high on the APQ. Later work (McFarland, 1975; Whitehead, Dresher, & Blackwell, 1976) showed that APQ scores are *not related* to accuracy of cardiac perception. Thus, it remains to be proven that accurate autonomic awareness is coupled to emotional experience and especially anxiety.

If we accept visceroperception as an integral part of emotional experience, the important question arises as to how the ability to perceive bodily changes is distributed in the population. It is certainly naive to expect this ability to be equally developed in different individuals; but if there are differences, they ought to appear in traits such as "strength of emotional experience," or "frequency of emotional experience." Obviously, this question is of special importance to clinical psychology: here quite often physiological measures are taken as indicators of the strength of emotional experience. But if different persons do not *perceive* processes to the same extent, objective changes in physiological measures provide only limited information about changes in subjectively experienced emotions, e.g. level of anxiety.

The bodily process which may be most easily accessible to *measurement* of visceral perception is heart activity, or more precisely, the occurrence of heart beats. On the one hand it is known that heart beats can be perceived at least to some degree by certain subjects, and on the other hand the actual occurrence of heart beats can be objectively detected, e.g. by EKG-recording. Additionally, by using the procedure described below for measuring perception of heart beats, one can quantify the accuracy of this aspect of perception of heart activity very easily. Besides, cardiac perception seems to be well suited to the study of the present question since

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there is no doubt that cardiac activity is associated with emotional experience.

Some studies have been conducted on perception of heart activity—most of them concerned with perception of heart *rate*—but none of them were done within the context of emotions. To the author's knowledge all of these studies investigated the relationship of heart rate perception to heart rate *control*. The literature dealing with this question is reviewed by Brener (1977), Carroll (1977) and Gannon (1977). Interestingly, in the studies using the APQ, no consistent correlation between APQ scores and objectively measured cardiac perception (or the ability to control heart rate) has been found.

In light of this, self-reports of visceral perception do not seem to be too closely associated with actual accuracy of viscerosception, but rather, they indicate what people *think* about the occurrence and intensity of bodily changes.

There seems to be no systematic study on the question of how the ability to perceive visceral changes is distributed; one reason for this is certainly the difficulty of measuring the accuracy of viscerosception. Besides gaining information on this topic, the aim of the present study was to test the following two hypotheses: a) subjects with good visceral perception report a higher level of emotional experience—in this case State Anxiety—than subjects with poor perception; b) on a more general level, it is hypothesized that a group of good perceivers will score higher on the personality trait "Emotional Lability" than a group of poor perceivers.¹

Method

Subjects

Forty subjects took part in this study; because of technical failure during one session one subject had to be discarded from the analyses. The age range of the remaining 39 subjects (16 female, 23 male) was 18–40 yrs. Twenty-nine were students, the rest were employed. None of them had taken part in a psychophysiological experiment before.

Experimental Task

The task consisted of the following: during a certain time interval, between onset and offset of a soft tone, presented by loudspeaker, the subject was to count his/her heart beats by concentrating on bodily feelings, possibly associated with the action of the heart. The subject was instructed *not* to take his/her own pulse or to try any other

physical manipulations which might facilitate the detection of heart beats. The task was performed 3 times, for time intervals lasting 25, 35 and 45 sec, respectively. After the termination of each tone the subject was requested to report the counted or estimated number of heart beats. Naturally, the subject was *not* informed how many seconds the 3 trials lasted.

Procedure

After the subject had arrived in the laboratory he/she was informed about the equipment, the different measures that were to be taken, and the general procedure of the experiment (but not yet about the specific experimental task). Then the electrodes were attached and the subject was seated in a reclining chair in the sound-damped and electrically shielded test chamber. The subject was then asked to complete the State-Trait Anxiety Inventory, Form X-1 (State Anxiety), by Spielberger, Gorsuch, and Lushene (1970). This was to be completed with regard to the present laboratory situation. Then the door to the adjacent room was closed, the subject was asked via loudspeaker to sit quietly during the whole session and, immediately after that, a 5-min baseline followed. At the end of this period the instruction explaining the perception task was given and the following series of alternating rest and perception periods started: rest (60 sec) - perception (25 sec) - rest (30 sec) - perception (35 sec) - rest (30 sec) - perception (45 sec) - rest (60 sec). Then the electrodes were removed, the subject left the experimental chamber, and the personality questionnaires were given.

Recording of Physiological Data

EKG, skin conductance, and respiration were monitored via a Beckman Dynograph Type R-411. EKG was recorded from Beckman biopotential electrodes (lead II) which were filled with Beckman electrode electrolyte. The signal was fed into the coupler 9806 A (gain: 1 mV/cm, time constant: 1 sec). A PDP-11/34 computer detected the R-waves and printed the number of heart beats for each rest and perception interval separately.

The skin conductance electrodes were also Beckman biopotential electrodes, filled with an isotonic paste (HELLIGE), and were attached on the palm of the non-dominant hand, about 5 cm apart. A constant current of 10 μ A was applied (GSR Coupler 9842) and a time constant of 10 sec was chosen. Respiration was monitored with a thermistor that was fastened to the top of a flexible wire. The wire was attached to a spectacle frame so that the thermistor could be positioned easily in front of a nostril without disturbing the subject. The Thermistor Coupler 9858 was used with a time constant of 1 sec.

All signals were recorded on paper at 2 mm/sec.

Measures and Analyses

Since the GSR coupler is not very suitable for direct measurement of conductance (because it is very insensitive to changes in base level, if constant voltage is applied), skin resistance was recorded (in the constant current mode) and transformed arithmetically to conductance units. The measures of interest were conductance level and

¹In addition to State Anxiety and Emotional Lability, a number of other personality variables, less related to emotions, were measured in this study; the results concerning these variables were reported elsewhere (Schandry, Note 1).

number of spontaneous fluctuations (SF) (amplitude criterion for SFs was $0.1 \mu\text{mho}$).

Respiration was analyzed only with regard to frequency.

The accuracy of perception of heart beats was quantified as an error score: the difference between reported and actual (as obtained from the EKG) number of heart beats was divided by the actual number of heart beats. The error score was obtained as the sum of the absolute values of the three quotients (from the three perception phases). Thus, a high error score reflected a poor accuracy of perception.

The personality characteristic "Emotional Lability" is one of the dimensions of the FPI (Freiburger Persönlichkeits-Inventar; Fahrenberg, Selg, & Hampel, 1978), which is a personality inventory widely used in Germany. It is a multidimensional personality test, with 212 items pertaining to general behavior, attitudes, habits, psychic condition, and bodily complaints. The questionnaire was developed on the basis of factor- and item-analytic procedures and has 12 scales. Normative data exist—separately for males and females—for three age ranges. The 12 scales are: nervousness, aggressiveness, depressiveness, excitability, sociability, calmness, striving for dominance, recsiveness, openness, extraversion, emotional lability (neuroticism), and masculinity. Validating studies have been conducted with the MPI and MMPI for instance, yielding high correlations for corresponding scales. Items loading high on the factor "emotional lability—neuroticism" are, for instance: "Sometimes I feel rather miserable, without any obvious reason," "In general I am quiet and it is rather difficult to make me excited" (negatively weighted), "My mood changes rather quickly," "I am sometimes morose and bad-tempered," and "Often I feel like a bomb just before exploding." Persons scoring high on the dimension "Emotional Lability" are described for instance, as experiencing rapidly changing moods, being excitable and distractable.

Actual emotional experience, in this case anxiety level, was investigated using the STAI-X1. The raw data from both questionnaires were transformed to T-scores prior to further statistical analyses.

Results

Influence of the Perception Task on the Physiological Measures

Table 1 shows the means over the rest and perception conditions, averaged over the 39 subjects. The perception task obviously had a pronounced

TABLE 1

Means and standard deviations of physiological measures during rest and task conditions

Conditions	Means (SDs in Parentheses)			
	HR (bpm)	SFs/min	SCL (μmhos)	Resp/min
Rest	61.1 (11.0)	5.9 (2.9)	25 (52)	11.8 (2.1)
Task	80.1 (16.8)	7.0 (4.9)	25 (52)	16.9 (3.1)

arousing effect, which is evident in the elevated means for heart rate, number of SFs, and respiration rate. (The unexpected high mean values in skin conductance seem to be due to 1 subject, who displayed an extremely elevated conductance level of $330 \mu\text{mhos}$).

Two-tailed *t*-tests for correlated means yielded the following *t*-values: heart rate, $t(38) = 22.6$, $p < .001$; SF, $t(38) = 2.11$, $p < .05$; conductance level, $t(38) = -0.03$; respiration rate, $t(38) = 15.2$, $p < .001$. For both heart rate and respiration rate each subject showed an increase in the mean values from rest to perception periods.

Distribution of Error Scores Within the Sample

Table 2 gives the error scores in ranked order, with the additional information of how often each subject achieved a "correct" result. "Correct" perception was scored when the counted number of heart beats differed by only ± 2 or less from the actual number of beats. A deviation of ± 2 was tolerated, since at tone onset and offset one beat can easily be missed or added. Figure 1 shows the frequency distribution of the error scores. The

TABLE 2
Error scores and number of correct results in the perception task for all subjects

Subject Positions in Ranked Order	Error Scores	No. of Correct Results ^a	Subject Positions in Ranked Order	Error Scores
1	0.000	3	23	0.771
2	0.019	3	24	0.781
3	0.022	2	25	0.823
4	0.109	2	26	0.975
5	0.122	2	27	1.017
6	0.129	2	28	1.129
7	0.211	2	29	1.146
8	0.305	2	30	1.173
9	0.352	1	31	1.241
10	0.376	1	32	1.296
11	0.387	1	33	1.355
12	0.392	2	34	1.442
13	0.395	—	35	1.606
14	0.410	1	36	1.703
15	0.420	1	37	1.821
16	0.463	1	38	1.902
17	0.523	—	39	2.247
18	0.544	1		
19	0.648	—		
20	0.709	—		
21	0.741	—		
22	0.749	1		

^aHow often a subject's count of heart beats differed by less than ± 2 from the actual number of heart beats. Subjects whose rank is above 22 never did achieve a correct result.

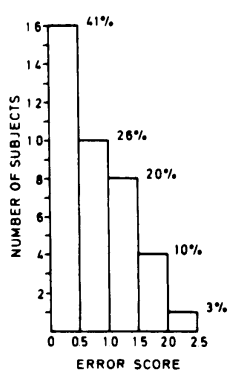


Figure 1. Distribution of error scores among the subjects, and percentages of the total number of subjects (39) scoring in each range.

range between the values 0.0 and 0.5 was achieved by 41% of the subjects. As Table 2 shows, all but 1 of the subjects who achieved a correct result at least once are found in this range.

The raw data were checked for training effects: no improvement from trial 1 to trial 3 was observable.

Formation of Extreme Groups

Two extreme groups were formed on the basis of visceral perception. The group of good perceivers (group G) included the 17 subjects achieving a correct result at least once in the perception task. The group of poor perceivers (group P) included the 17 subjects with the highest error scores (right column in Table 2). The possibility of testing group G against the rest of the subjects was discarded, since the remaining subjects included some with an error score lower than the poorest perceiver of group G (but not fulfilling the requirement of at least one correct result).

Surface Validity of the Perception Task

The possibility exists that the members of the good group might have achieved their better results by just counting accidentally at the correct rhythm, but without actually perceiving their heart beats. If this were the case, their counting rate should be con-

stant over the 3 trials; that is, the difference between counted and measured heart beats should exhibit a high variability in those subjects with a markedly varying heart rate and a low variability in subjects with stable heart rates over trials. In order to check for this, the correlation (Pearson's *r*) between standard deviations of heart rates and standard deviations of differences was computed across the 17 subjects of the good group. A small, positive, but nonsignificant correlation (*r* = .24) was obtained. The low correlation between the two variabilities suggests that in this group counting rate followed to some extent changes in heart rate.

Comparisons Between Extreme Groups

Table 3 shows the means of the physiological measures over rest and task phases for the two extreme groups. There is virtually no difference for heart rate and respiration rate between the two groups, a small difference in number of SFs, but a pronounced difference in conductance level. (As inspection of the raw data revealed, however, this difference seems to be due mainly to 1 subject in group P having a conductance level as low as 1.5 μ mhos and another subject in group G displaying a value as extreme as 330 μ mhos).

Table 4 shows the means, standard errors and *t*-values of group differences in the means for the variables State Anxiety and Emotional Lability. The means of both variables are significantly higher for good perceivers than for poor perceivers.

Discussion

The perception task was accompanied by marked changes in three of the four physiological indicators. The changes in heart rate, number of spontaneous fluctuations, and respiration rate seem to demonstrate an increase in autonomic arousal during the task. Skin conductance level remained constant, which may be a consequence of the rather rapid changes between rest and perception phases; possibly, this tonic measure was not sufficiently responsive to rapid changes, so that the mean values remained unchanged. The increase in respiration rate in all subjects during the perception task shows

TABLE 3

Means and standard deviations of physiological measures for the two groups during rest and task conditions

Groups	Means (SDs in Parentheses)							
	HR (bpm)		SFs/min		SCL (μ mhos)		Resp/min	
	Rest	Task	Rest	Task	Rest	Task	Rest	Task
Good	60.7 (11.9)	79.8 (17.4)	6.8 (3.1)	8.2 (5.7)	36 (77)	36 (77)	11.9 (1.9)	16.9 (2.5)
Poor	60.9 (12.9)	80.0 (17.2)	5.0 (2.1)	6.3 (4.6)	18 (18)	18 (18)	11.7 (2.5)	16.8 (4.0)

TABLE 4

Means and standard errors of emotional lability and state anxiety measures and results of t-tests for group differences between good and poor perceivers

Emotional Lability			State Anxiety		
Means (S.E. in Parentheses)			Means (S.E. in Parentheses)		
Good Perceivers	Poor Perceivers	t	Good Perceivers	Poor Perceivers	t
53.6 (1.9)	47.6 (2.4)	1.99*	54.6 (1.6)	48.1 (1.5)	2.88**

* $p < .05$.

** $p < .01$.

that nobody tried to improve their perception of heart beats by slowing the breathing rate or by holding his/her breath. Similar to this study, Porges and Raskin (1969) found an increase in heart rate (though *phasic* heart rate) during a task requiring perception of heart activity. The observed acceleration in the present study is *not*, however, interpreted as a correlate of rejection of environmental input (Lacey, e.g. 1967), in contrast to the interpretation by Porges and Raskin (1969) of *their* data. I view the heart rate increase, together with the increase in respiration rate and number of SFs, as an indication of general sympathetic arousal due to the task demands.

The distribution of the accuracy of heart beat perception within the sample shows that perfect perception appeared quite infrequently. Only 2 subjects out of 39 always reached the criterion of correct perception. The number of subjects with relatively good, but not perfect, perception was much higher. On the other hand, there were nearly as many subjects with an almost total lack of information about the occurrence of their heart beats; inspection of Table 2 shows that 13 subjects had an error score above 1, which means that they missed the actual number of heart beats on the average with an error of 33%. Obviously, the ability to perceive heart beats is distributed over an extremely wide range, from absolutely precise perception to virtually no perception at all, at least during conditions of physical inactivity.

It is important to note that there was no difference in mean heart rate between the two groups of good and poor heart beat perceivers. Therefore, it is unlikely that the better perception of the good group was a consequence of higher heart rate level or reactivity.

Furthermore, the group of good perceivers, although scoring significantly higher on State Anxiety, did not differ from the others in heart rate, respiration rate, or the number of spontaneous fluctuations (the means of skin conductance level are

not considered because of the 2 subjects with the extreme conductance levels). Obviously, higher anxiety was *not* accompanied by elevated autonomic activity in the present data.

We do not know much about the processes mediating the perception of heart beats (anecdotal data from this study suggest that heart beats are primarily perceived in the chest and to a lesser degree in the neck and head). For instance, it cannot be ruled out that good cardiac perception was facilitated by an elevated blood pressure level for the members of the group of good perceivers, the higher blood pressure in turn being a concomitant of the higher state anxiety observed in this group. This would mean that the relationship between cardiac awareness and emotional experience could be a consequence of higher blood pressure reactivity to emotion-provoking situations. But, as long as no data on the influence of blood pressure level on cardiac perception are available, these considerations remain speculative.

In light of the present data it seems that higher self-reported anxiety is due to better perception of physiological processes rather than to actual level of autonomic arousal. Naturally, this should have some consequences on the treatment of physiological data as "indicators of anxiety." In clinical psychology, heart rate is one of the most widely used physiological measures; often it is assumed that changes in this variable reflect changes in the emotional experience, for instance anxiety. This may be true, but as the present results suggest, *perception* of heart activity seems to be a more important factor than absolute heart rate or heart rate changes.

Concordant with the above considerations, the group of good perceivers also scored higher on the personality trait "Emotional Lability." If a subject is regularly aware of heart activity, situations of potential emotional content will lead to comparatively more frequent emotional experiences (of any kind).

Summing up, the following conclusions seem to follow from the results of the present study: a) emo-

tional experience is linked to perception of bodily processes; b) the accuracy of perception of visceral changes varies over a wide range among individuals; c) observed changes in a certain autonomic function

of an individual do not necessarily index changes in his emotional experience, as long as there is no information about that individual's ability to perceive visceral processes.

REFERENCES

- Brener, J. Visceral perception. In J. Beatty & H. Legewie (Eds.), *Biofeedback and behavior*. New York: Plenum, 1977. Pp. 235-259.
- Carroll, D. Cardiac perception and cardiac control: A review. *Biofeedback and Self-Regulation*, 1977, 2, 349-369.
- Fahrenberg, J., Selg, H., & Hanpel, R. *Das Freiburger Persönlichkeitsinventar, FPI*. Göttingen: Verlag für Psychologie, Dr. C. J. Hogrefe, 1978.
- Gannon, L. The role of interoception in learned visceral control. *Biofeedback and Self-Regulation*, 1977, 2, 337-347.
- James, W. What is an emotion? *Mind*, 1884, 9, 188-205.
- Lacey, J. I. Somatic response patterning and stress: Some revisions of activation theory. In M. H. Appley & R. Trumbull (Eds.), *Psychological stress: Issues in research*. New York: Appleton-Century-Crofts, 1967. Pp. 14-42.
- Mandler, G., Mandler, J. M., & Uviller, E. T. Autonomic feedback: The perception of autonomic activity. *Journal of Abnormal and Social Psychology*, 1958, 56, 367-373.
- McFarland, R. A. Heart rate perception and heart rate control. *Psychophysiology*, 1975, 12, 402-405.
- Porges, S. W., & Raskin, D. C. Respiration and heart rate components of attention. *Journal of Experimental Psychology*, 1969, 81, 497-503.
- Spielberger, C. D., Gorsuch, R. L., & Lushene, R. E. *STAI manual for the State-Trait Anxiety Inventory*. Palo Alto: Consulting Psychologists Press, Inc., 1970.
- Whitehead, W. E., Drescher, V. M., & Blackwell, B. Lack of relationship between Autonomic Perception Questionnaire scores and actual sensitivity for perceiving one's heart beat. *Psychophysiology*, 1976, 13, 177. (Abstract)

REFERENCE NOTE

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